Introduction to Kubernetes

What is Kubernetes?

A **Production-Grade Container Orchestration System** Google-grown, based on Borg and Omega, systems that run inside of Google right now and are proven to work at Google for over 10 years.

- Extend the container goodness across nodes
- Recovers automatically from failure
- Basic monitoring, logging, health checking
- Enables containers to find each other.
- Reduce cost to run many things in production. Enables new ways of building applications.

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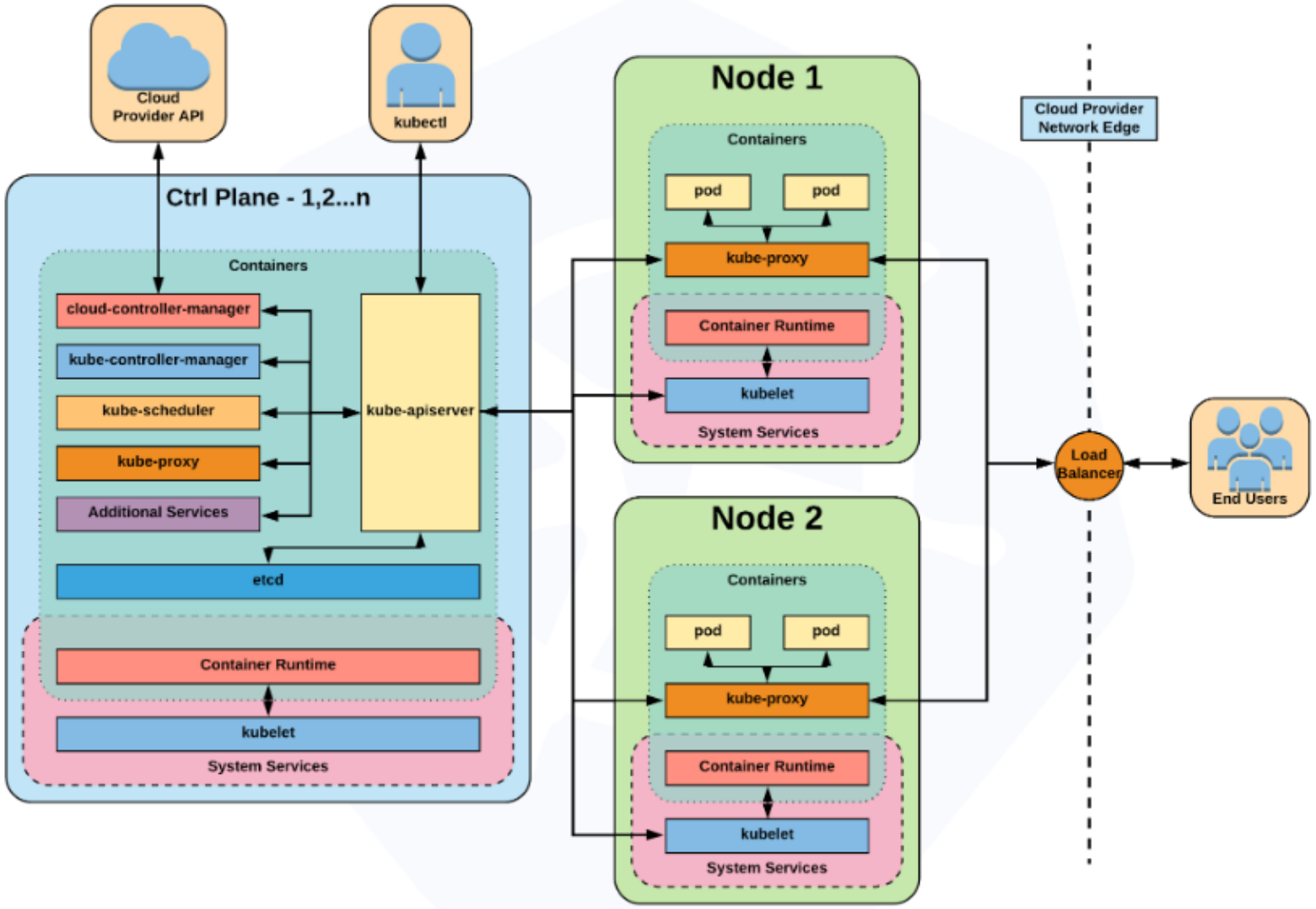
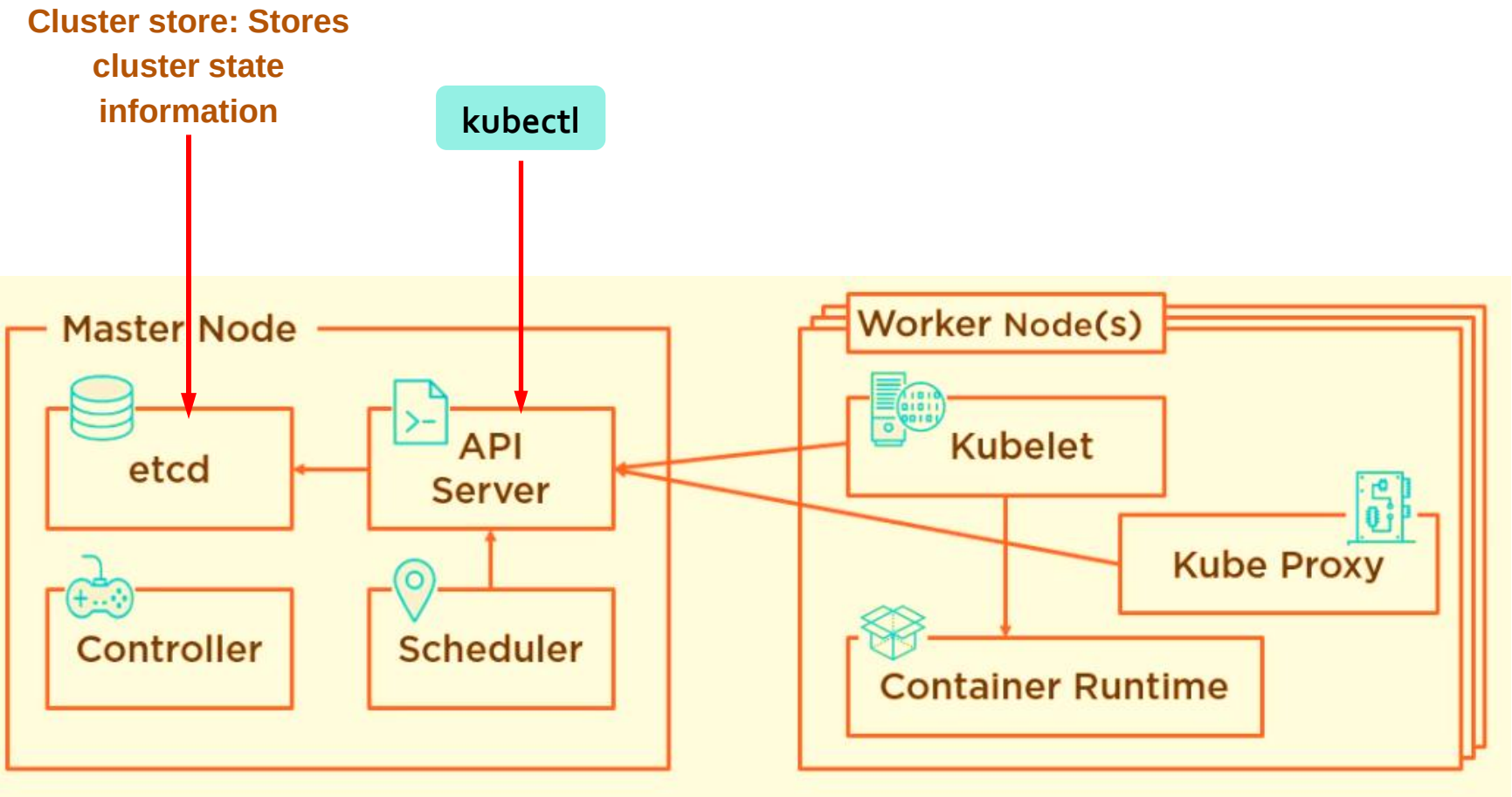
What Kubernetes can do ?

- Self Healing
- Autoscale Workloads
- Deployments - Roll Forwards / Rollbacks
- Fire off jobs and scheduled cronjobs
- Manage Stateless and Stateful Applications
- Provide native methods of service discovery
- Easily integrate and support 3rd party apps

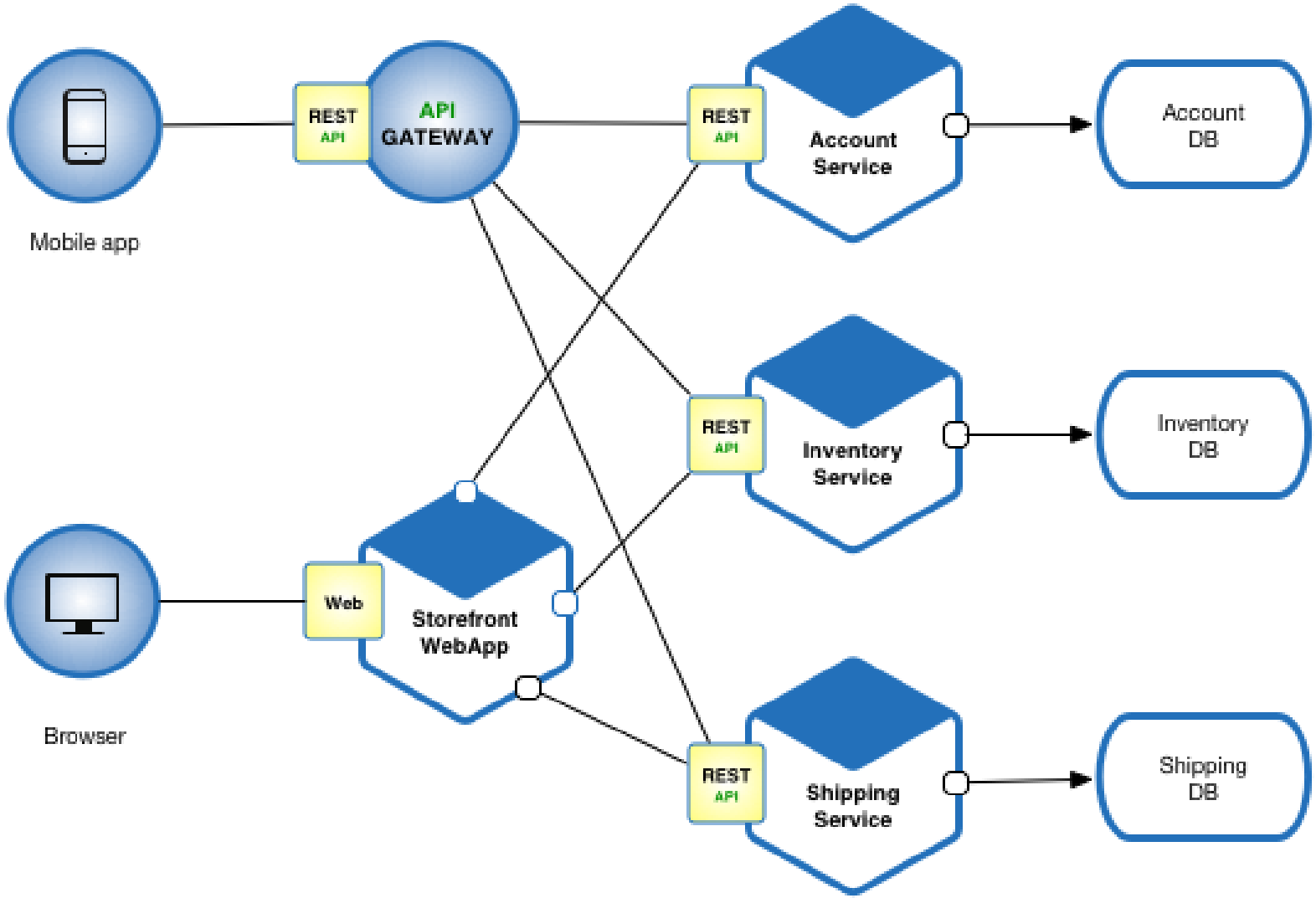
Kubernetes Architecture



Kubernetes Architecture



Micro service applications



Kubernetes Cloud services

Elastic Kubernetes Service (EKS)

https://aws.amazon.com/getting-started/projects/deploy-kubernetes-app-amazon-eks/

Google Kubernetes Engine (GKE)

https://cloud.google.com/kubernetes-engine/docs/how-to/

Azure Kubernetes Services (AKS)

https://docs.microsoft.com/en-us/azure/aks/kubernetes-walkthrough

Azure Kubernetes Service (AKS)

- Azure Kubernetes Service (AKS) simplifies deploying a managed Kubernetes cluster in Azure by offloading the operational overhead to Azure. As a hosted Kubernetes service, Azure handles critical tasks, like health monitoring and maintenance.
- Built-in Auto-scaling and Load Balancing and CI/CD with GitHub, Docker and Azure container registry.

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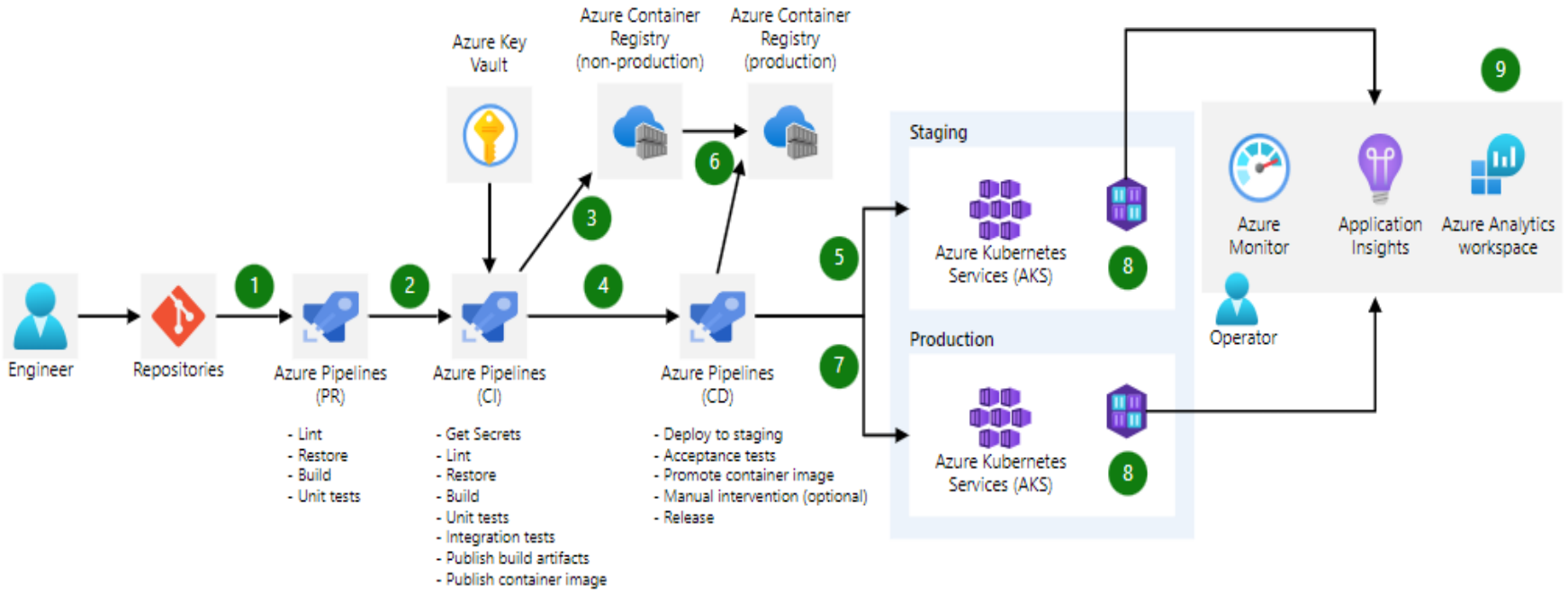
CI/CD for AKS Apps with Azure Pipeline



Microsoft Azure

Comparison

Hypervisor / Virtualization

VMs

(Atomic Unit of Scheduling)

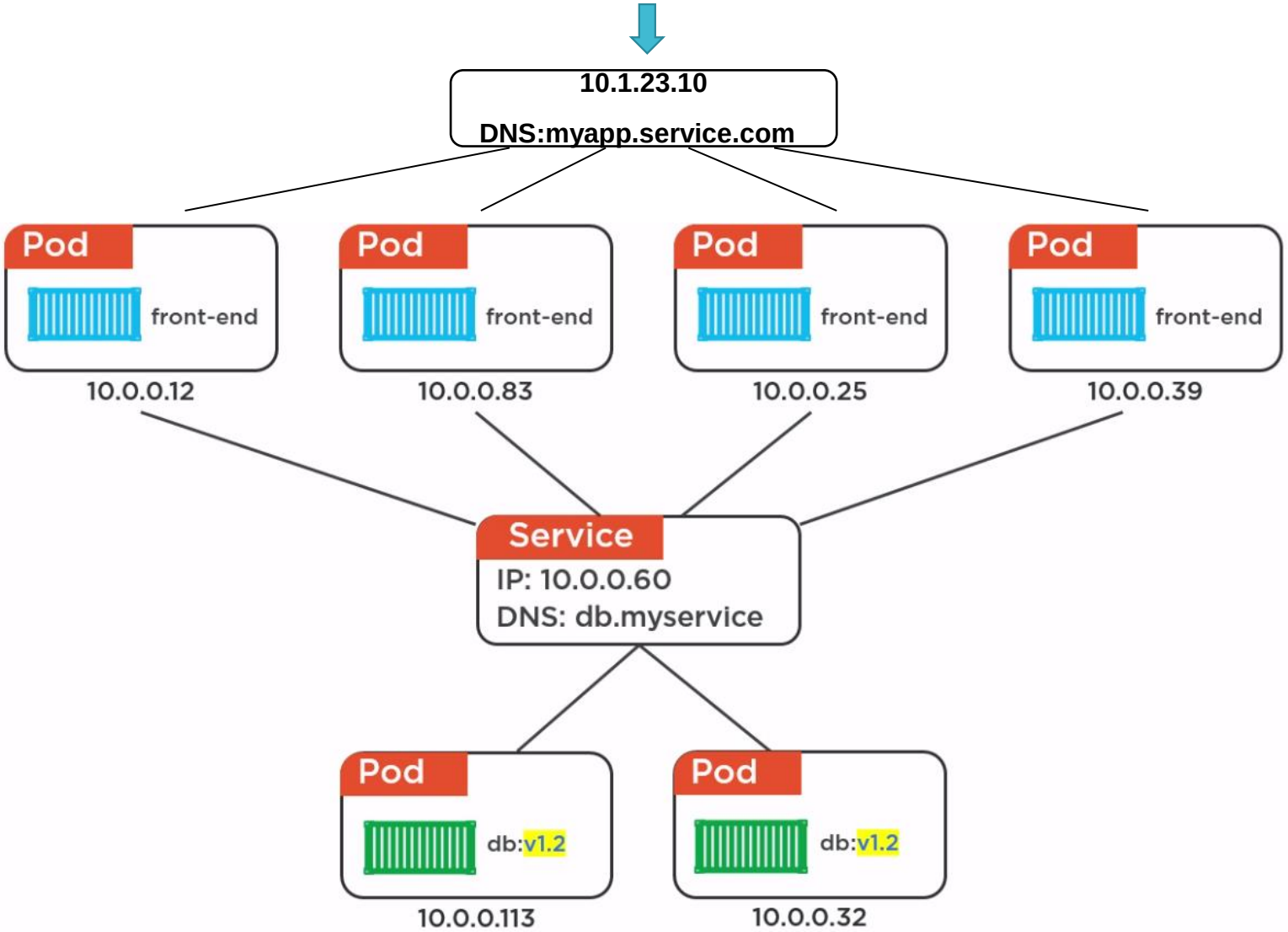
Docker

Containers

(Atomic Unit of Scheduling)

Kubernetes

PODs

(Atomic Unit of Scheduling)

Load Balancer Services